

July 22, 2026

DPF Meeting 2026 @ Fermilab

Proton TMDPDFs from Lattice QCD

Spin Dependence of the Non-perturbative Transverse Structure

Jinchen He

In collaboration with ANL-BNL group



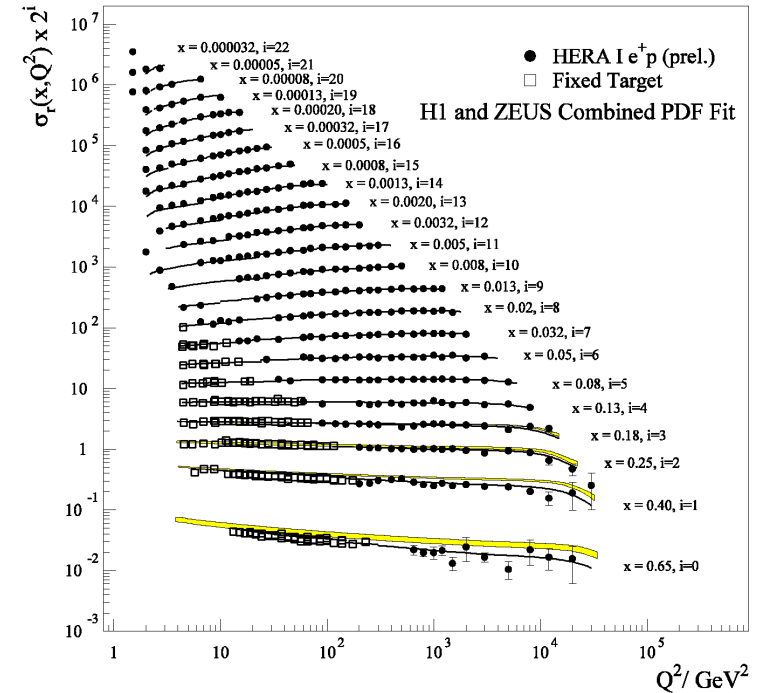
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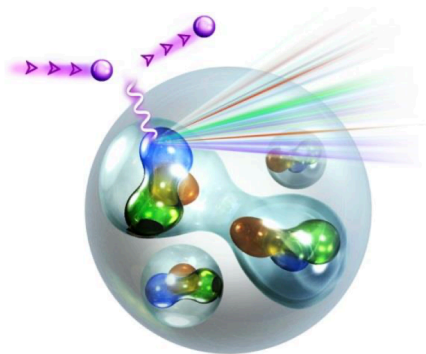
Parton Physics: Collinear structures

Bjorken scaling in 1968

- The structure functions are approximately independent of Q^2
- Evidence for point-like structures
- Parton model! *R. P. Feynman, Conf. Proc. C 690905 (1969)*
- Scaling violation at large $Q^2 \rightarrow$ running coupling in QCD



Collinear factorization in QCD



Cr. Dave Gaskell

$$\sigma_{\text{DIS}} \propto \left| \begin{array}{c} l \\ l' \\ q \\ P \end{array} \right|^2 \approx \left| \begin{array}{c} k \approx xP \\ p \end{array} \right|^2 \otimes \left| \begin{array}{c} l \\ l' \\ q \\ xP \end{array} \right|^2$$

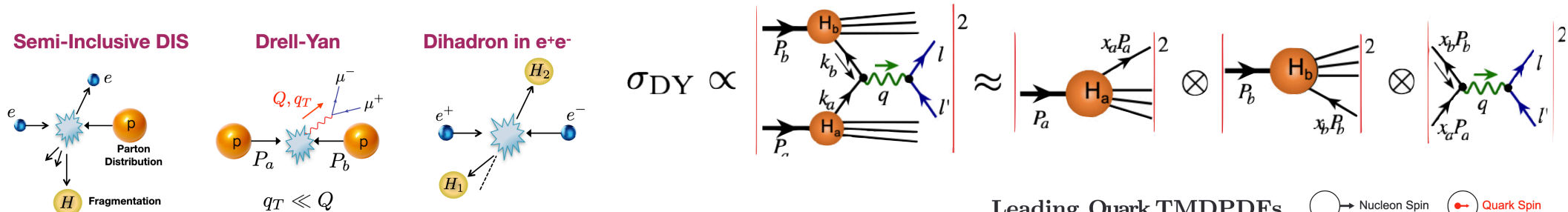
PDF
Pert. Scattering

R. Boussarie, et al., "TMD Handbook", [arXiv:2304.03302 [hep-ph]]

Parton Physics: Collinear to TMD

TMD processes and TMD factorization

- TMD processes are important processes in high energy collisions, like Drell-Yan process on LHC and Semi-Inclusive DIS on EIC;



- But why we need TMD factorization? $\left\{ \begin{array}{l} \text{Scale separation} \\ \text{Large log: } \ln(Q/q_T) \\ \text{Soft mode} \end{array} \right.$

- When $q_T \sim Q$, collinear factorization is enough

$$\frac{d\sigma}{d^4q} = \sum_{i,j} \int_{x_a}^1 d\xi_a \int_{x_b}^1 d\xi_b \boxed{f_{i/H_a}(\xi_a)} f_{j/H_b}(\xi_b) \frac{d\hat{\sigma}_{ij}(\xi_a, \xi_b)}{d^4q} \left[1 + O\left(\frac{\Lambda_{\text{QCD}}^2}{q_T^2}, \frac{\Lambda_{\text{QCD}}^2}{Q^2}\right) \right]$$

- When $\Lambda_{\text{QCD}} \lesssim q_T \ll Q$, TMD factorization is needed

$$\frac{d\sigma}{d^4q} = \frac{1}{s} \sum_{i \in \text{flavors}} \hat{\sigma}_i^{\text{TMD}}(Q) \int d^2\mathbf{k}_T \boxed{f_{i/H_a}(x_a, \mathbf{k}_T)} f_{i/H_b}(x_b, \mathbf{q}_T - \mathbf{k}_T) \left[1 + O\left(\frac{q_T^2}{Q^2}, \frac{\Lambda_{\text{QCD}}^2}{Q^2}\right) \right]$$

Leading Quark TMDPDFs $\bigcirc$ Nucleon Spin $\bigcirc$ Quark Spin

		Quark Polarization		
		Un-Polarized (U)	Longitudinally Polarized (L)	Transversely Polarized (T)
Nucleon Polarization	U	$f_1 = \bigcirc$ Unpolarized		$h_1^\perp = \bigcirc - \bigcirc$ Boer-Mulders
	L		$g_1 = \bigcirc - \bigcirc$ Helicity	$h_{1L}^\perp = \bigcirc - \bigcirc$ Worm-gear
	T	$f_{1T}^\perp = \bigcirc - \bigcirc$ Sivers	$g_{1T}^\perp = \bigcirc - \bigcirc$ Worm-gear	$h_1 = \bigcirc - \bigcirc$ Transversity $h_{1T}^\perp = \bigcirc - \bigcirc$ Pretzelosity

R. Boussarie, et al., 2304.03302 (2023)

✚ Phenomenological Extractions of TMDs

Significant progress has been made in the phenomenological parameterizations of TMDs

- Collins-Soper kernel (CS kernel): rapidity evolution kernel of TMDs; Some of groups incorporate neural network;
Some of groups include lattice data;
A. Bacchetta, et al. (MAP) JHEP 08 (2024); V. Moos, et al., JHEP 11 (2025) ...

- Pion TMDs

- Unpolarized *A. Vladimirov, JHEP 10 (2019); M. Cerutti, et al. (MAP), Phys.Rev.D 107 (2023);
P. C. Barry, et al. (JAM) Phys.Rev.D 108 (2023)*

- Nucleon TMDs

- Unpolarized

M. Bury, et al., JHEP 10 (2022); A. Bacchetta, et al., JHEP 10 (2022); V. Moos, et al., JHEP 05 (2024); A. Bacchetta, et al., JHEP 08 (2024); V. Moos, et al., JHEP 11 (2025); P. Barry et al., 2510.13771 ...

- Helicity and Transversity

L. Gamberg, et al., PRD 106 (2022); A. Bacchetta, et al., PRL 134 (2025); ...

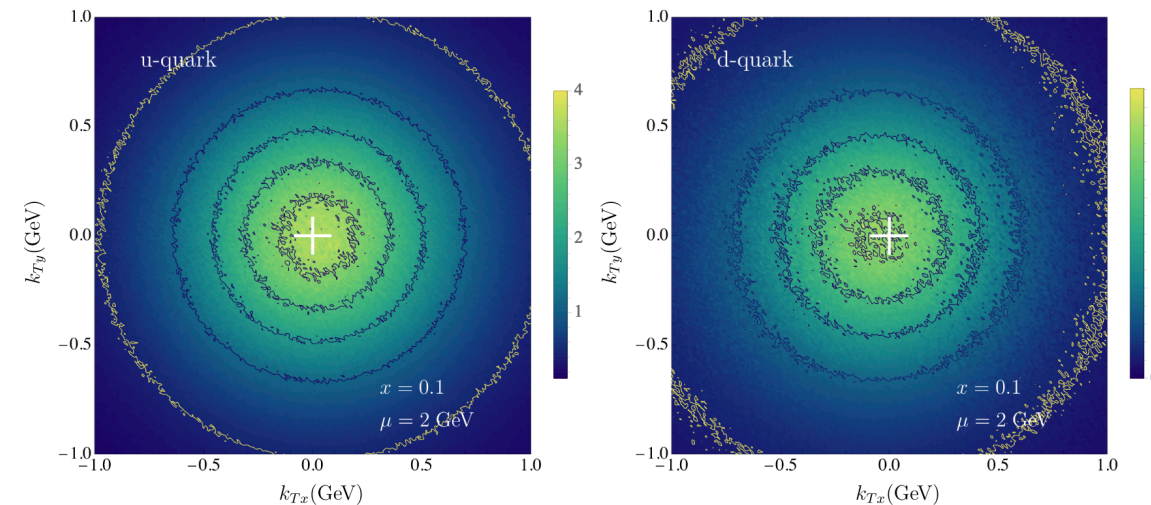
- Sivers

M. Bury, et al., PRL 126 (2021); I. P. Fernando, et al., Phys.Rev.D 108 (2023) ...

- Boer-Mulders

Z. Lu, et al., Phys.Rev.D 81 (2010); X. Liu, et al., Eur.Phys.J.C 81 (2021) ...

Tomographic scan of the nucleon via quark density function





Lattice Calculations of TMDs

Lattice QCD provides first-principles predictions of TMDs

- Mellin Moments; *B.U. Musch, et al., Phys.Rev.D 85 (2012); B. Yoon, et al., 1601.05717; B. Yoon, et al., Phys. Rev. D 96 (2017) ...*
- Large Momentum Effective Theory (LaMET)
 - CS kernel *M. H. Chu, et al. (LPC), JHEP 08 (2023); A. Avkhadiev et al., PRL 132 (2024); D. Bollweg, et al., Phys. Lett. B 852 (2024); J-X Tan, et al. (LPC), 2511.22547; A. Avkhadiev et al., 2510.26489, A. Francis et al., 2606.19221 ...*
 - Intrinsic soft functions *Q. A. Zhang, et al. (LPC), PRL 125 (2020); M. H. Chu, et al. (LPC), JHEP 08 (2023)*
 - Unpolarized TMD *JH, et al. (LPC), Phys.Rev.D 109 (2024); D. Bollweg, JH, et al., Phys.Rev.D 106 (2022)*
 - Helicity TMD *D. Bollweg, X. Gao, S. Mukherjee and Y. Zhao, PRL 135 (2025)*
 - Boer-Mulders TMD *L. Walter, et al. (LPC), PRD 111 (2025); L. Ma, et al. (LPC), JHEP 08 (2025)*



Large-Momentum Effective Theory (LaMET)

TMDPDF is defined from a light-cone correlator in a hadron, which is boost invariant.

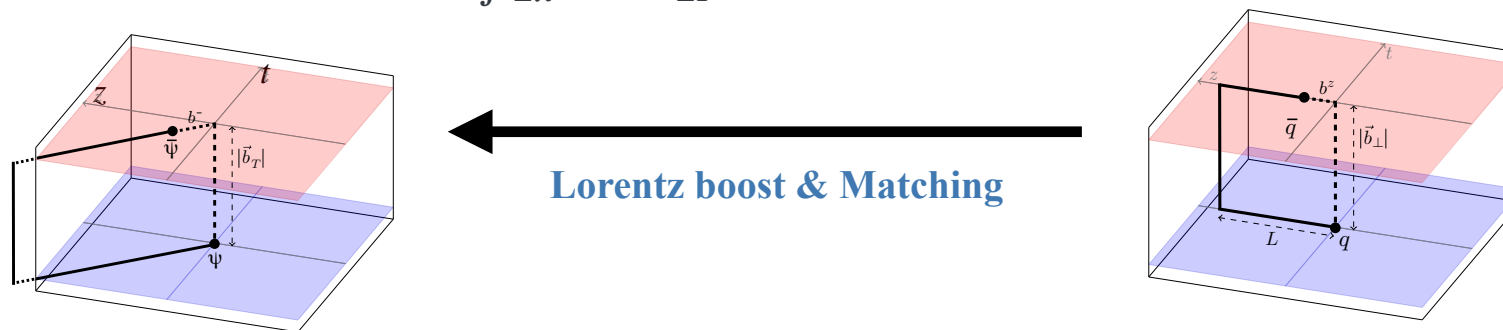
$$f(x, b_{\perp}, \dots) = \int_{-\infty}^{\infty} \frac{db^-}{2\pi} e^{-ib^-(xP^+)} \left\langle P \left| \bar{\psi}(b^{\mu}) W_{\square}(b^{\mu}, 0) \frac{\gamma^+}{2} \psi(0) \right| P \right\rangle \longleftrightarrow \left\langle |\vec{P}| = \infty \left| O(t=0) \right| |\vec{P}| = \infty \right\rangle$$

Equal-time correlator! Parton model!

Define a quasi distribution with large-momentum states and time-independent operators.

$$\tilde{f}_{\Gamma}^0(x, b_{\perp}, P^z, \mu) = P^z \int \frac{dz}{2\pi} e^{iz(xP^z)} \frac{1}{2P^t} \langle P | \bar{\psi}_0(z, b_{\perp}) W_{\square}(z, b_{\perp}; 0) \Gamma \psi_0(0) | P \rangle$$

Differ by UV structures



LaMET enables us to obtain the precision-controlled TMD PDFs in moderate x region.

$$\sqrt{S_I(b_{\perp}; \mu)} \cdot \tilde{f}_{\Gamma}(x, b_{\perp}, P^z; \mu) = f(x, b_{\perp}; \mu, \zeta) H_f(x, P^z; \mu) \exp \left[\frac{1}{2} \ln \frac{(2xP^z)^2}{\zeta} \gamma^{\overline{\text{MS}}}(b_{\perp}; \mu) \right] + \text{Power corrections}$$



Soft gluon radiation will lead to existence of soft functions

$$\sqrt{S_I(b_\perp; \mu)} \cdot \tilde{f}_T(x, b_\perp, P_z; \mu) = f(x, b_\perp; \mu, \zeta) H_f(x, P_z; \mu) \exp \left[\frac{1}{2} \ln \frac{(2xP^z)^2}{\zeta} \gamma^{\overline{\text{MS}}}(b_\perp; \mu) \right]$$

After regularization, the rapidity evolution is controlled by Collins-Soper scale: $\zeta = 2(xP^+)^2 e^{-2y_n}$

- Due to the factorization of the soft mode from the collinear mode, the soft function contains the well-

known rapidity divergence;

$$\underbrace{\int_{q_T}^Q \frac{dk}{k}}_{\text{full}} = \lim_{\tau \rightarrow 0} \left[\underbrace{\int_0^Q \frac{dk}{k} R_c(k, \tau)}_{\text{collinear}} + \underbrace{\int_{q_T}^\infty \frac{dk}{k} R_s(k, \tau)}_{\text{soft}} \right] = \ln \frac{Q}{q_T}$$

- The soft function can be separated into two parts:

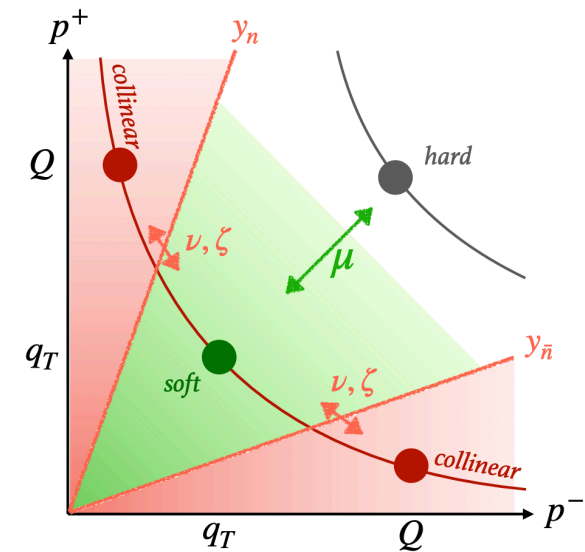
- Rapidity evolution kernel: CS kernel $\gamma^{\overline{\text{MS}}}(b_\perp; \mu)$

- Rapidity independent part: intrinsic soft function $\sqrt{S_I(b_\perp; \mu)}$

- CS kernel can be extracted from the rapidity evolution of TMDs

$$\tilde{\phi}_{\tilde{\Gamma}}(x, b_T, P_z; \mu) = A(x, b_T) H_\phi(x, \bar{x}, P_z; \mu) \exp \left[\ln \frac{P_z}{P_{\text{fix}}} \tilde{\gamma}(x, b_T) \right]$$

Here we do a fit on quasi-TMDWF at various pion momenta



R. Boussarie, et al., 2304.03302 (2023)



Soft Functions

Extract intrinsic soft functions from large-momentum form factors

- The intrinsic soft function cannot be directly calculated on lattice because of two light-like

Wilson lines in different directions;

- Fortunately, it can be extracted from the meson form factor

$$F(b_{\perp}, P_1, P_2, \Gamma, \Gamma') \equiv -4N_c \frac{\langle P_2 | \bar{q}(b_{\perp}) \Gamma q(b_{\perp}) \bar{q}(0) \Gamma' q(0) | P_1 \rangle}{f_{\pi}^2(P_1 \cdot P_2)}$$

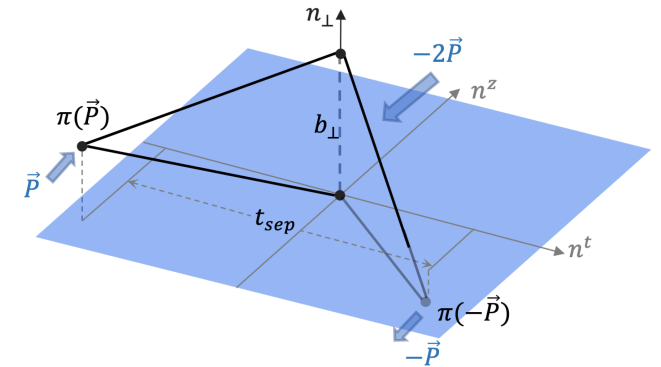
X. Ji, Y. Liu and Y. S. Liu, Nucl.Phys.B 955 (2020)

- The form factor satisfies the factorization formula

$$F(b_{\perp}, P^z) = \int dx_1 dx_2 H_F(x_1, x_2, P^z; \mu) \phi^{\dagger}(x_1, b_{\perp}, y_n; \mu, \zeta_1, \bar{\zeta}_1) \phi(x_2, b_{\perp}, -y_n; \mu, \zeta_2, \bar{\zeta}_2)$$

- Therefore, the intrinsic soft function can be extracted via

$$S_I(b_{\perp}; \mu) = \frac{F(b_{\perp}, P^z)}{\int dx_1 dx_2 H_F(x_1, x_2, P^z; \mu) \tilde{\Phi}^{\dagger}(x_1) \tilde{\Phi}(x_2)} \text{ where } \tilde{\Phi}(x) \equiv \frac{\tilde{\phi}_{\Gamma}(x, b_{\perp}, P^z; \mu)}{H_{\phi}(x, \bar{x}, P^z; \mu)}$$



Q. A. Zhang, et al., Phys. Rev. Lett. 125 (2020)

Now we are ready to extract the TMDPDFs



Coulomb Gauge Method

Define a quasi distribution in CG without Wilson line

$$\tilde{f}_{\text{CG}}^0(x, b_{\perp}, P^z, \mu) = P^z \int \frac{dz}{2\pi} e^{iz(xP^z)} \frac{1}{2P^t} \langle P | \bar{\psi}_0(z, b_{\perp}) \Gamma \psi_0(0) | P \rangle \Big|_{\vec{\nabla} \cdot \vec{A} = 0}$$

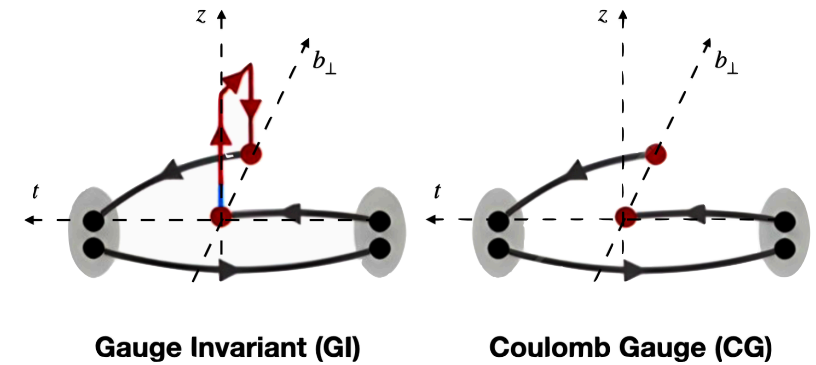
Y. Zhao, PRL 133 (2024)

X. Gao, W. Y. Liu and Y. Zhao, PRD 109 (2024)

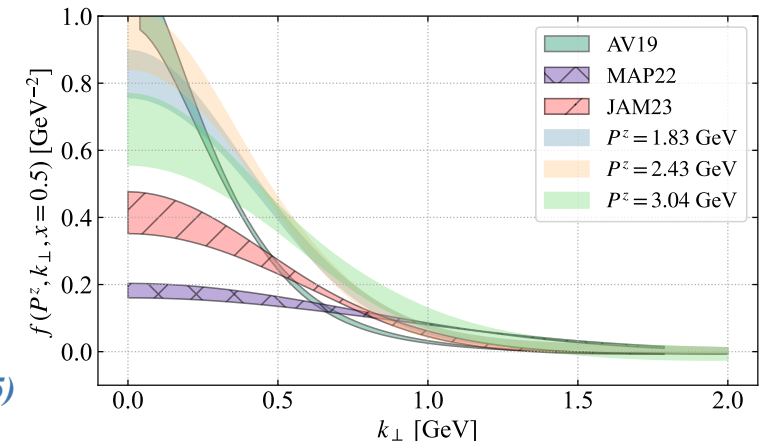
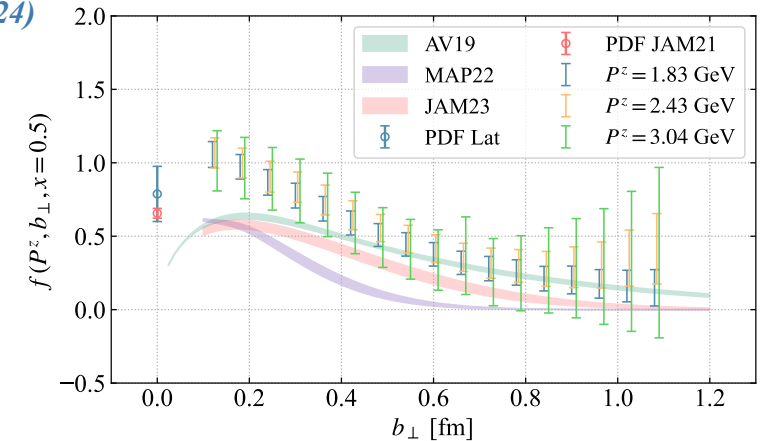
Why choose CG?

- The quasi distribution in CG belongs to the universality class in LaMET;
- No self-energy of the Wilson link, improving the SNR significantly;
- Simplified renormalization: $\bar{\psi}_0(z, b_{\perp}) \Gamma \psi_0(0) = Z_{\psi}(a) [\bar{\psi}(z, b_{\perp}) \Gamma \psi(0)]$;
- Larger off-axis momenta (3D rotational symmetry);
- A growing number of successful applications.

D. Bollweg, X. Gao, JH, S. Mukherjee and Y. Zhao, PRD 112 (2025)



Pion unpolarized TMD PDFs





Lattice Setup

Extract intrinsic soft functions from large-momentum form factors

- 2+1-flavor **domain-wall** fermion (for both sea and valence quark) with Iwasaki gauge action;

Chiral fermion with strongly suppressed operator mixing

- **Physical pion mass** (for both sea and valence quark);

D. B. Kaplan, Phys.Lett.B 288 (1992);

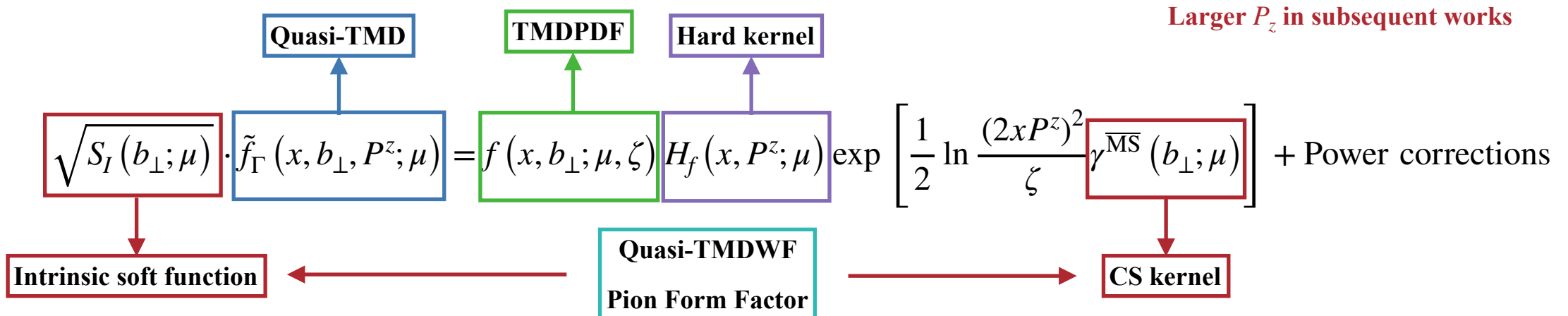
T. Blum and Y. Shamir, 2603.27608

- Lattice spacing is $a = 0.0836$ fm;

- The proton quasi-TMD beam functions computed at proton momentum $P_z = 1.62$ GeV;

- The pion quasi-TMD wave functions with pion momenta up to $P_z = 1.85$ GeV;

- Pion form factors at momentum transfers up to $Q^2 = 13.7$ GeV².



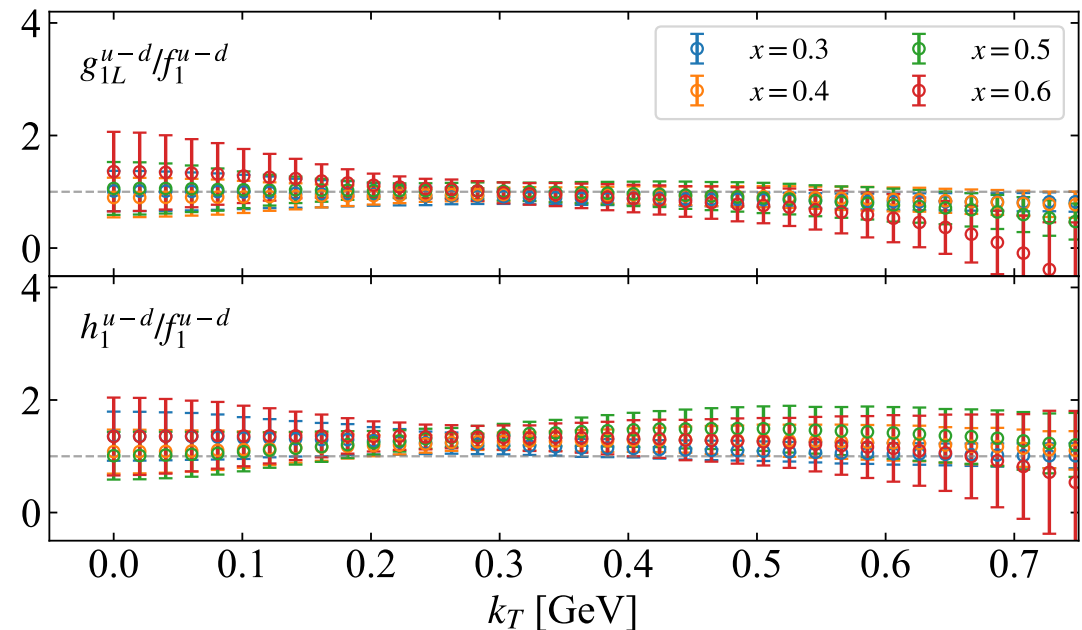
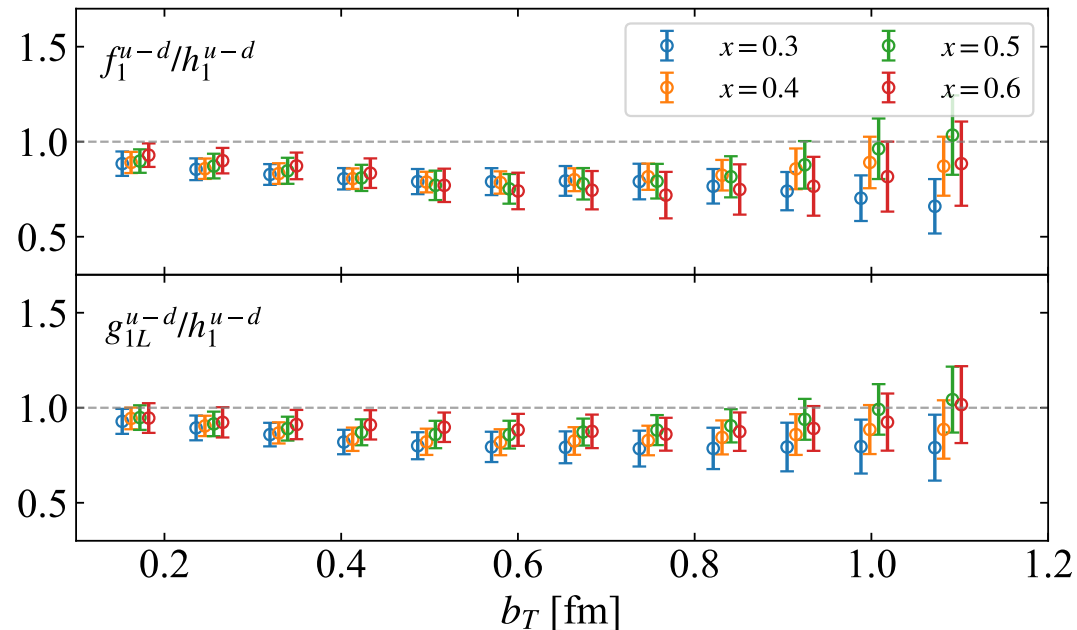


RGI Ratio of TMD PDFs

Ratio of TMD PDFs equals to the ratio of corresponding quasi-TMD beam functions

- This ratio is free from renormalization scheme and scale dependence;
- Thanks to the multiplicative matching, the ratio is independent from CS kernel, intrinsic soft functions;
- The non-perturbative transverse structure is largely insensitive to spin polarizations;
- The spin information primarily encoded in the longitudinal structure;

Consistent with the results in D. Bollweg et al., PRL 135 (2025)





Renormalization

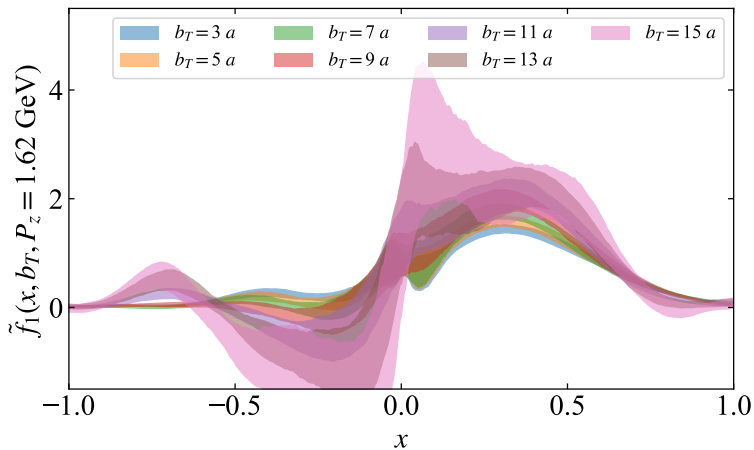
After renormalization, the universal large b_T behavior is mainly determined by the soft function

- The bare matrix elements of quasi-TMD beam functions and quasi-TMDWFs are renormalized as:

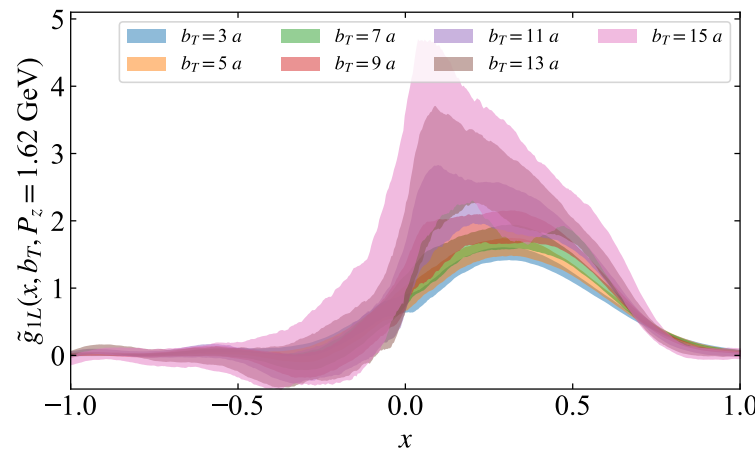
$$\tilde{h}_{\bar{\Gamma}}(b_z, b_T, P_z; \mu) = \frac{\tilde{h}_{\bar{\Gamma}}^0(b_z, b_T, P_z; a)}{\tilde{\varphi}_{\gamma^* \gamma^5}^0(0, b_T, 0; a)} \text{ and } \tilde{\varphi}_{\bar{\Gamma}}(b_z, b_T, P_z; \mu) = \frac{\tilde{\varphi}_{\bar{\Gamma}}^0(b_z, b_T, P_z; a)}{\tilde{\varphi}_{\gamma^* \gamma^5}^0(0, b_T, 0; a)};$$

- In the moderate x region, quasi-TMD beam functions in all three polarizations show minor b_T -dependence;

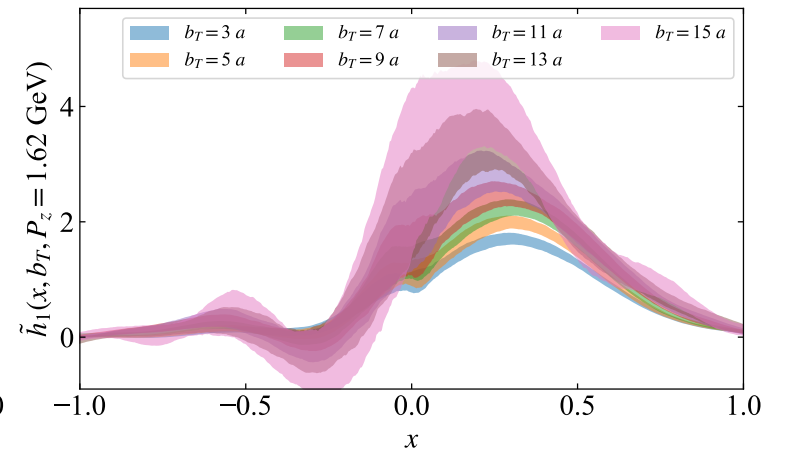
Unpolarized quasi-TMD



Helicity quasi-TMD



Transversity quasi-TMD



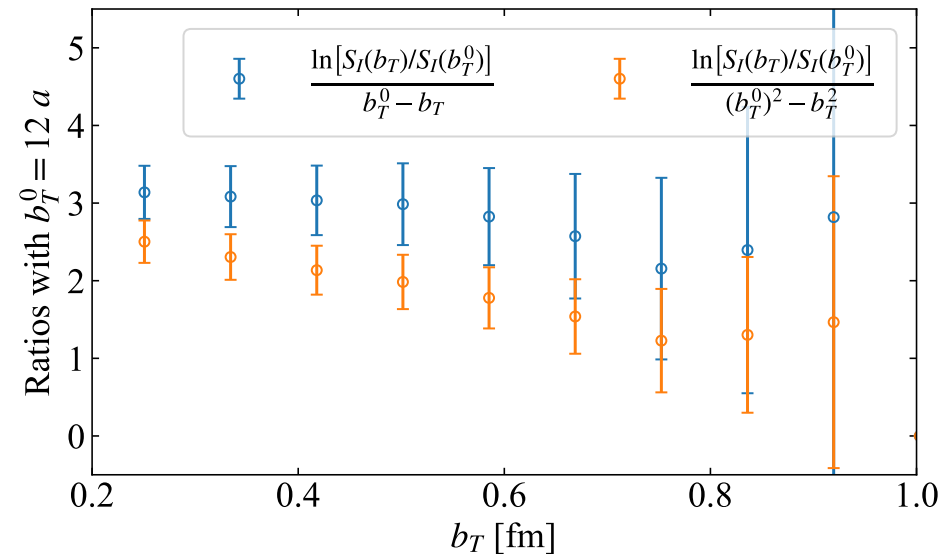
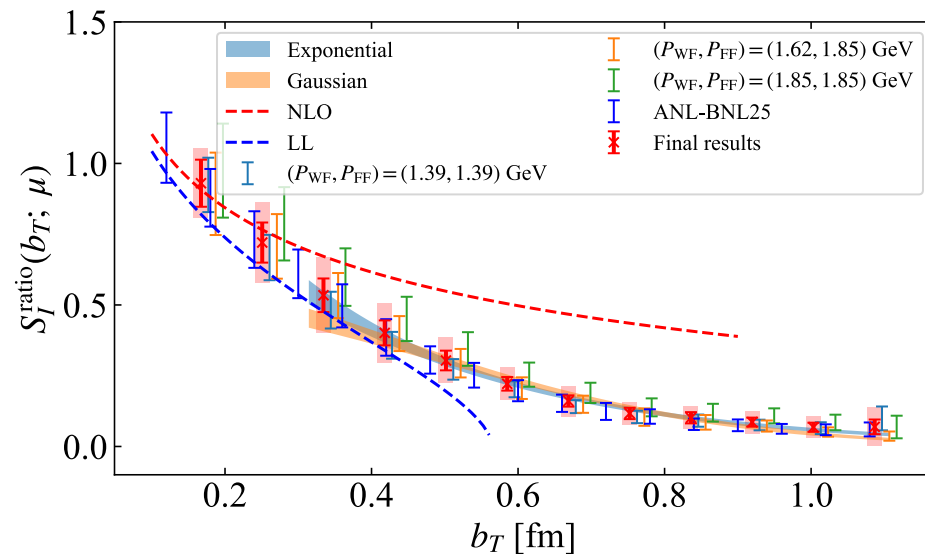


Intrinsic Soft Functions

Extract intrinsic soft functions at NLL from quasi-TMDWF and pion form factors

$$S_I(b_\perp; \mu) = \frac{F(b_\perp, P^z)}{\int dx_1 dx_2 H_F(x_1, x_2, P^z; \mu) \tilde{\Phi}^\dagger(x_1) \tilde{\Phi}(x_2)}$$

- Pion form factors at momentum transfers up to $Q^2 = 13.7 \text{ GeV}^2$;
- Large- b_T behavior: exponential (Ae^{-mb_T}) or gaussian ($Ae^{-m^2 b_T^2}$)?
- Fit results give $\chi^2/\text{d.o.f.} = 0.3$ v.s. 1.4 and $\log\text{GBF} = 20.2$ v.s. 14.2 (Bayes Factor of 400).



Effective mass plot also favors the exponential form

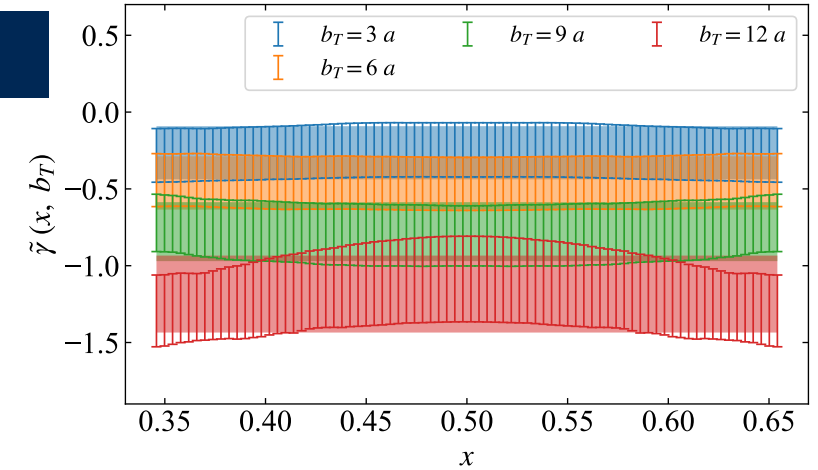


CS kernel

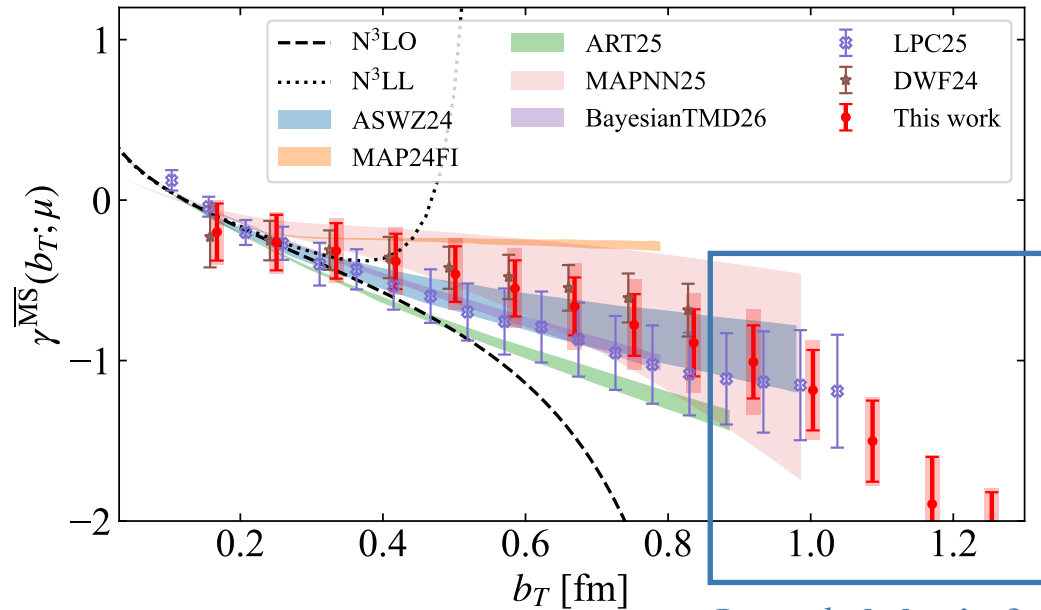
Extract CS kernel from quasi-TMDWF at different P_z

$$\tilde{\phi}_{\tilde{F}}(x, b_T, P_z; \mu) = A(x, b_T) H_{\phi}(x, \bar{x}, P_z; \mu) \exp \left[\ln \frac{P_z}{P_{\text{fix}}} \tilde{\gamma}(x, b_T) \right]$$

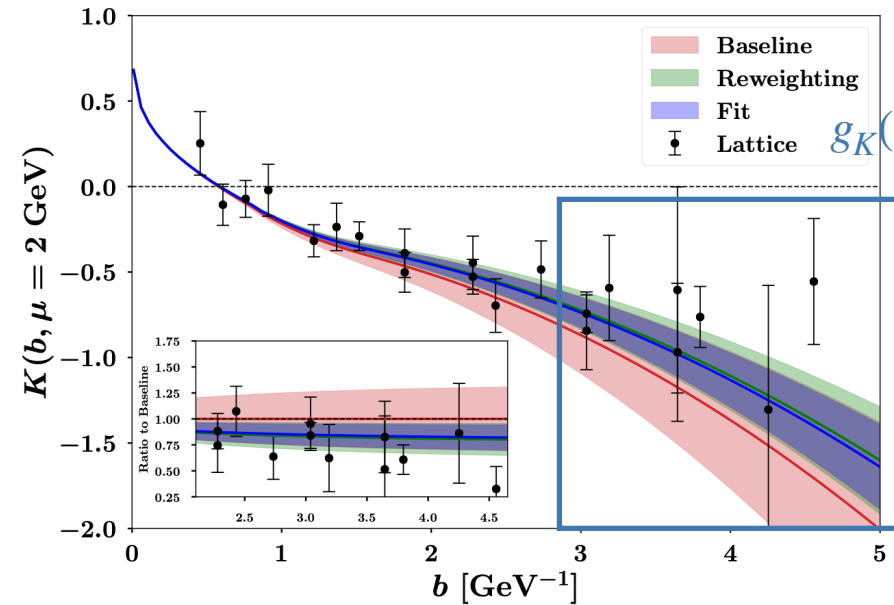
- Constant fits on $\tilde{\gamma}(x, b_T)$ for each b_T to get $\gamma^{\overline{\text{MS}}}$;
- Two sets of momenta are considered to estimate systematic error;



Good consistency with DWF24



Large b_T behavior?



A. Avkhadiev et al., Phys.Rev.Lett. 136 (2026)

$$g_K(b) = 2g_2^2 b^2$$

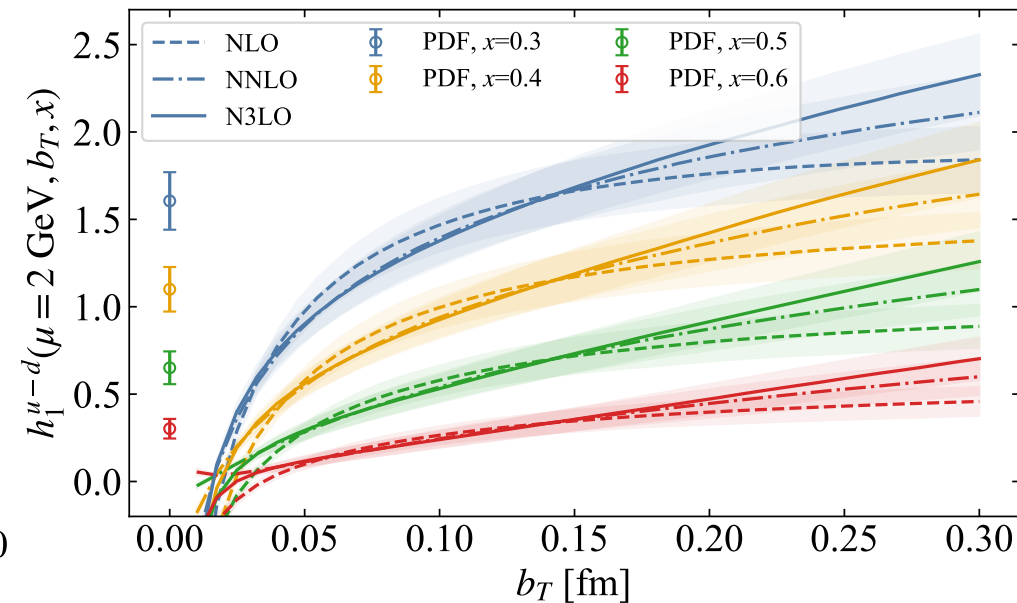
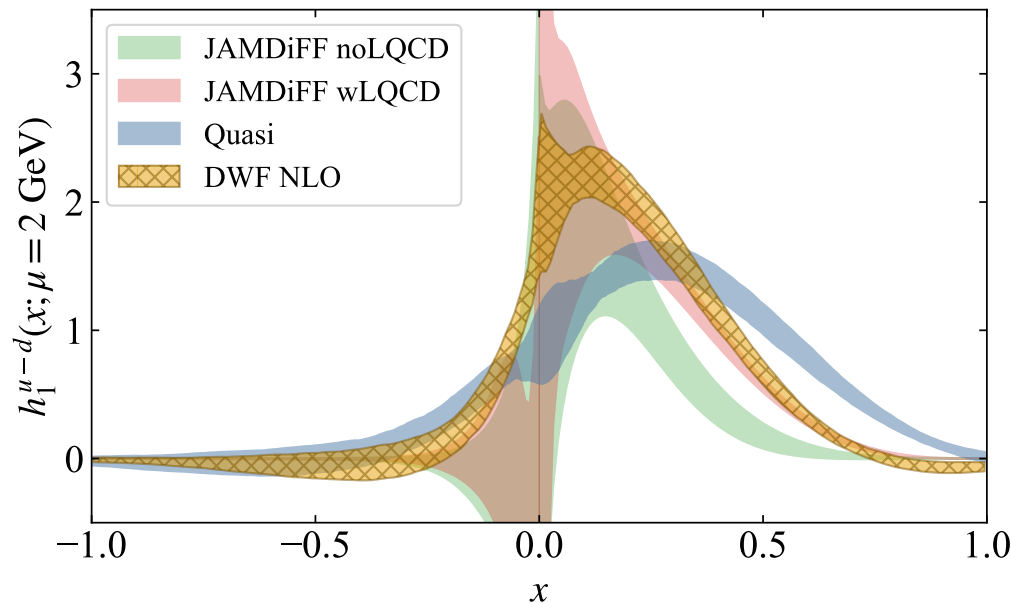


Short distance OPE

TMD PDFs admit an OPE onto the corresponding collinear PDFs at perturbative b_T

- The collinear PDFs can be extracted from correlators at $b_T = 0$, renormalized in hybrid scheme;
- OPE prediction is given as $f_{\Gamma}(x, b_T; \mu, \zeta) = C_{\Gamma}(x, b_T; \mu, \zeta) \otimes f_{\Gamma}^{\text{coll}}(x, \mu) + \mathcal{O}(b_T^2 \Lambda_{\text{QCD}}^2)$;
- The short distance matching kernels are available up to N3LO, showing a good convergence;

M.-X. Luo et al., JHEP 06 (2021) 115; Y. Zhu, JHEP 02 (2026) 115; Y. Zhu, JHEP 03 (2026)

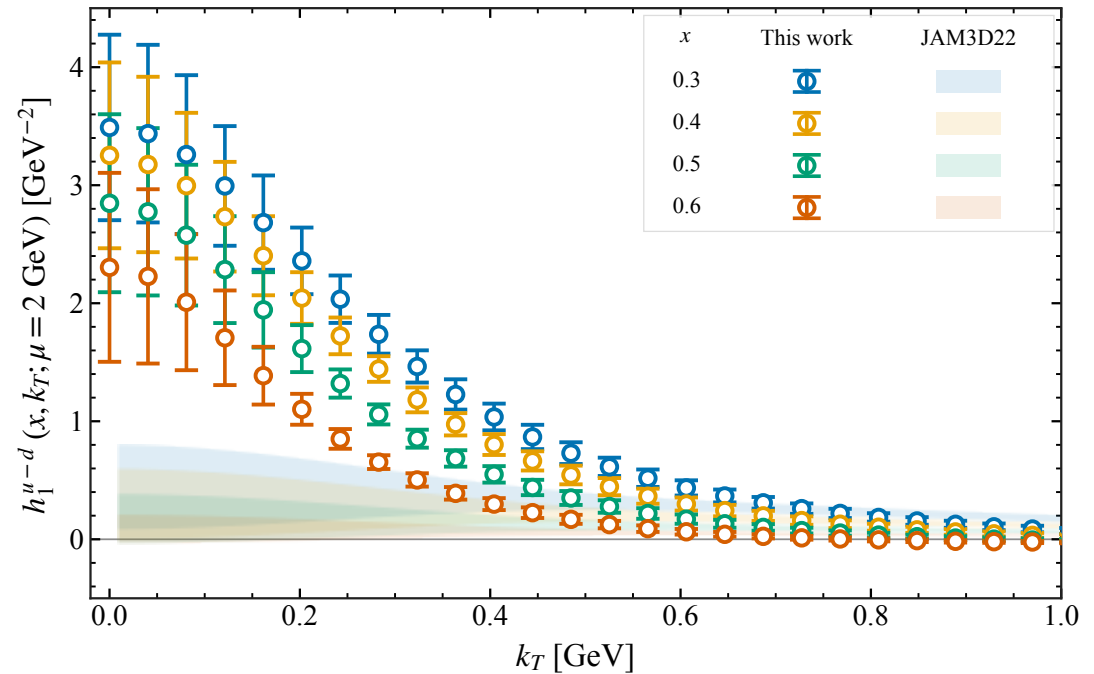
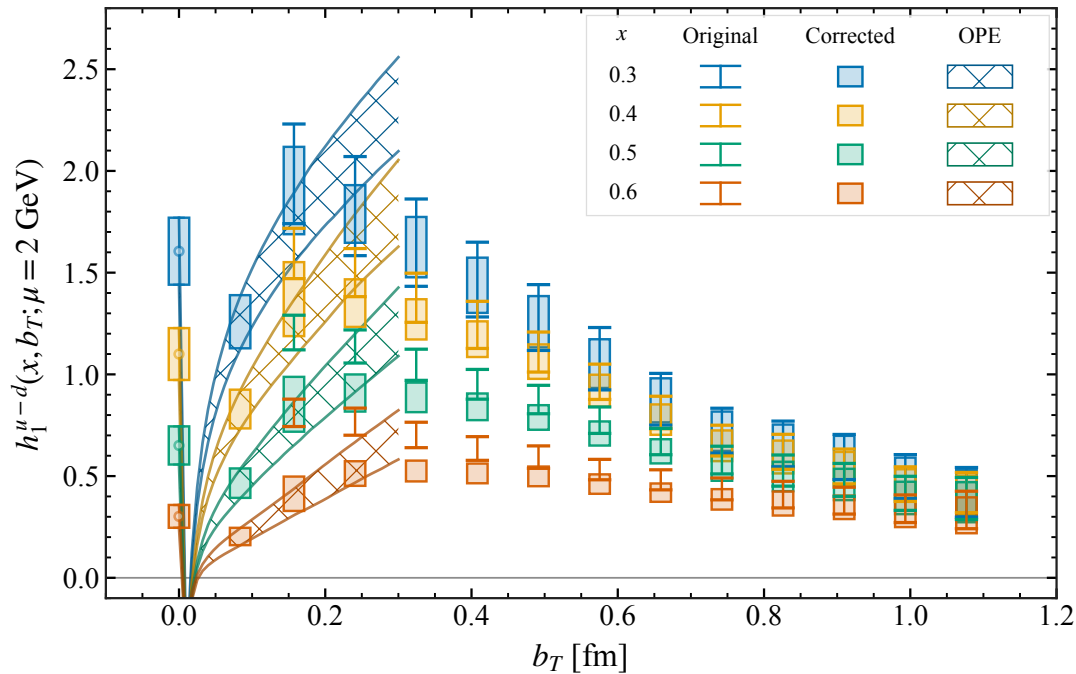




Transversity TMD PDFs

TMD PDFs in transverse coordinate and momentum space

- The short b_T region is corrected by a $\delta C(x, b_T, P_z) = 1 + c/(2xb_TP_z)$ to match OPE results;
- The large b_T region is extrapolated using exponential form (Ae^{-mb_T}) with loose prior from soft functions;

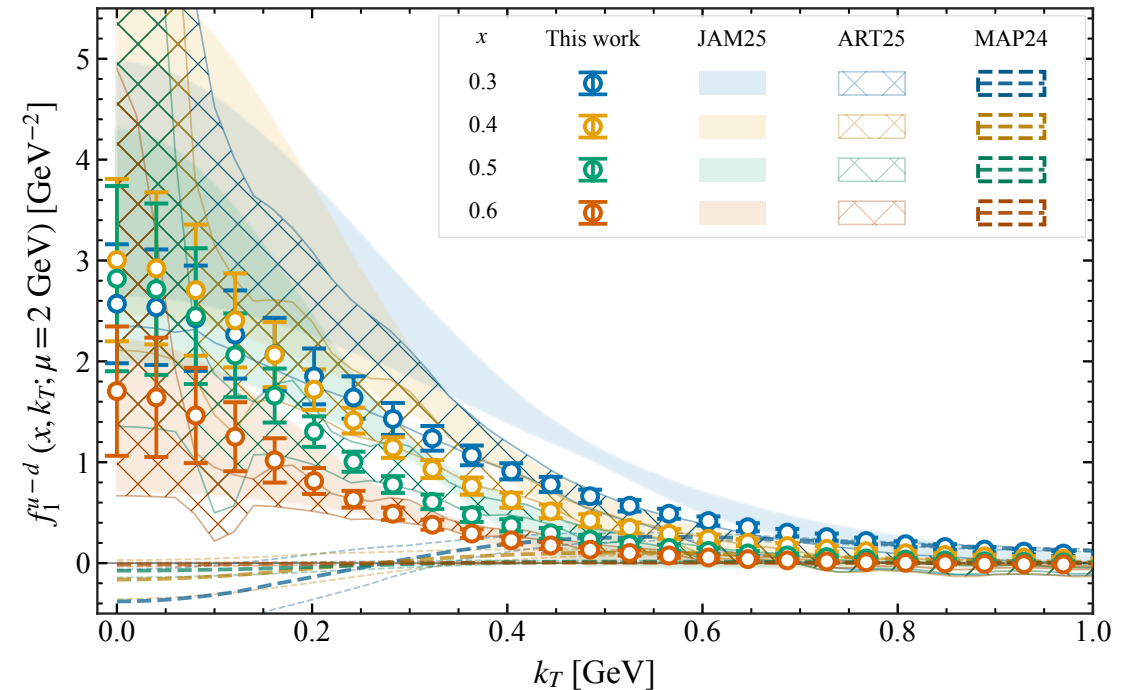
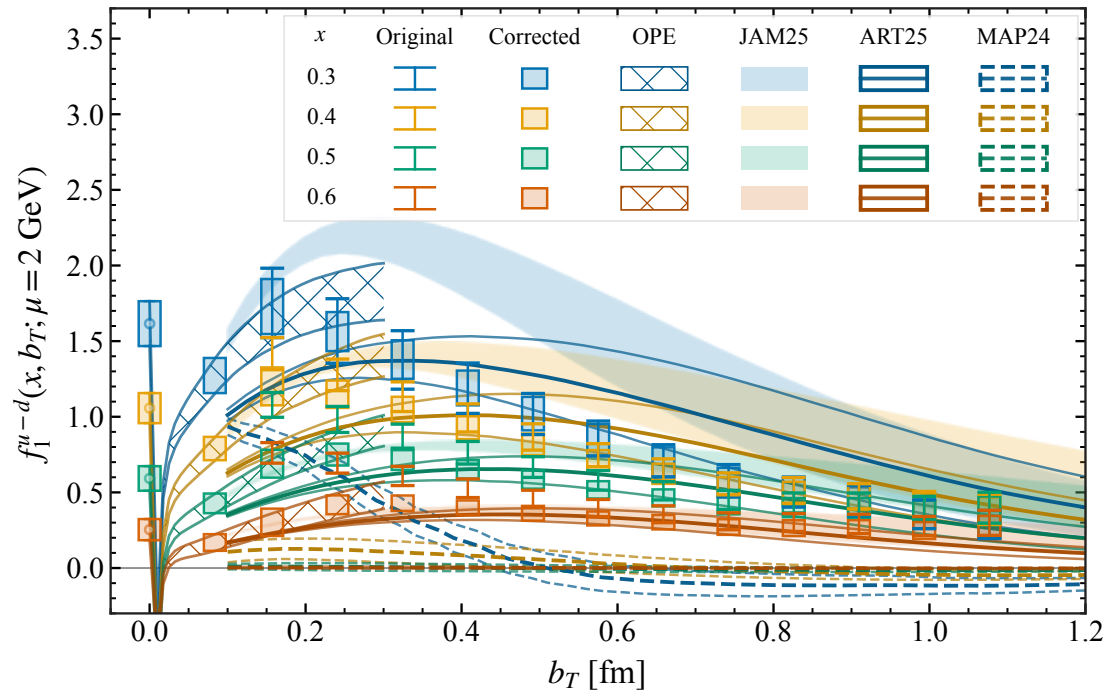




Unpolarized TMD PDFs

TMD PDFs in transverse coordinate and momentum space

- The short b_T region is corrected by a $\delta C(x, b_T, P_z) = 1 + c/(2xb_TP_z)$ to match OPE results;
- The large b_T region is extrapolated using exponential form (Ae^{-mb_T}) with loose prior from soft functions;

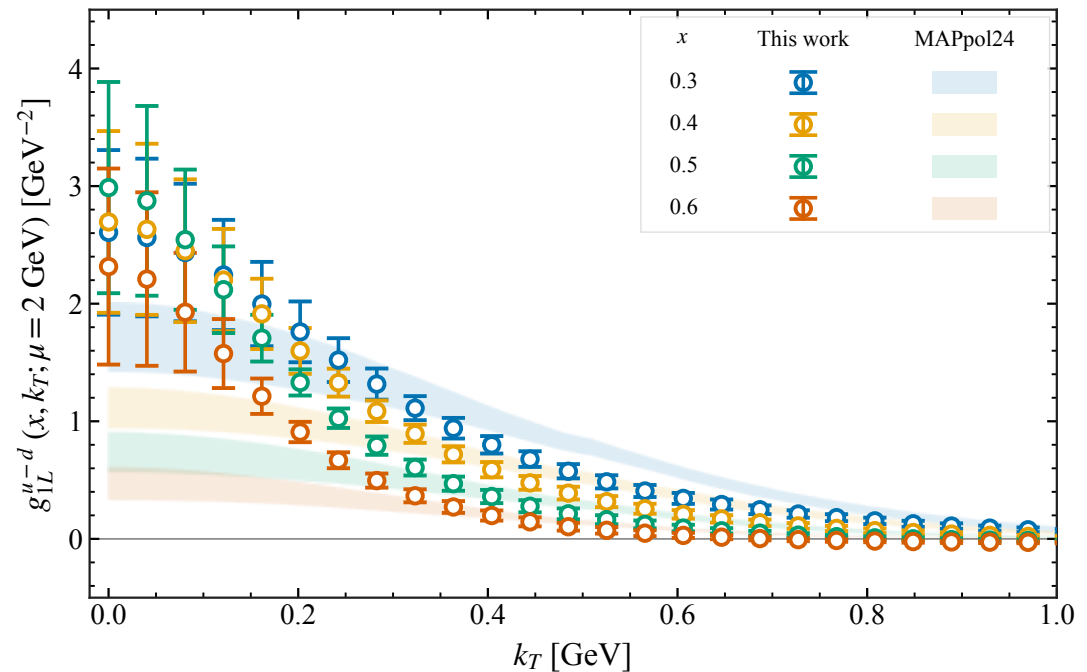
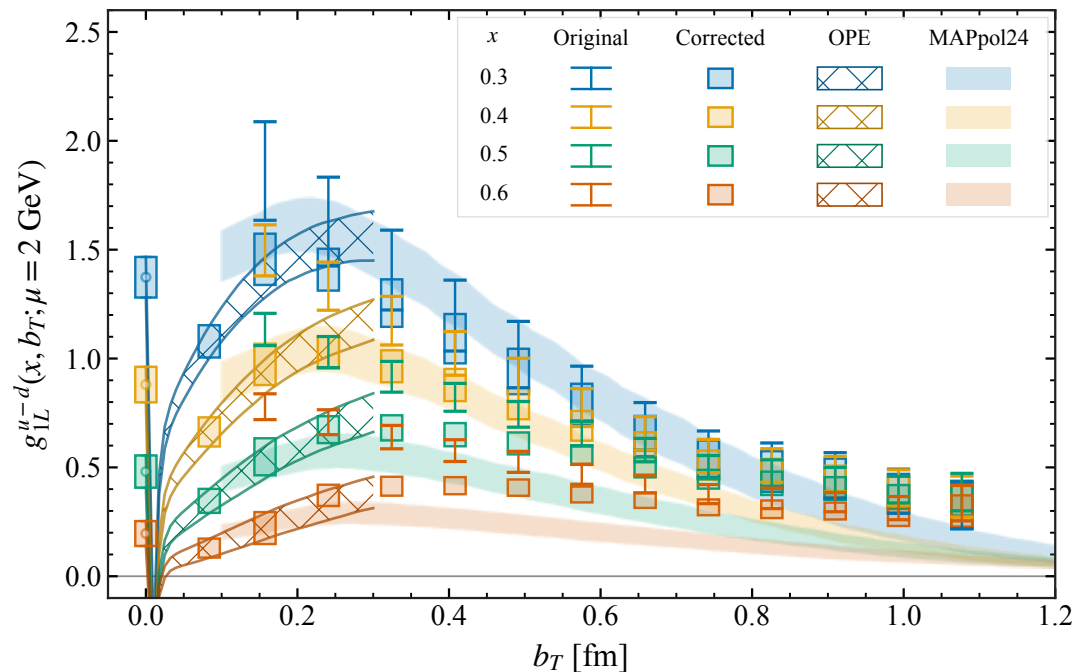




Helicity TMD PDFs

TMD PDFs in transverse coordinate and momentum space

- The short b_T region is corrected by a $\delta C(x, b_T, P_z) = 1 + c/(2xb_TP_z)$ to match OPE results;
- The large b_T region is extrapolated using exponential form (Ae^{-mb_T}) with loose prior from soft functions;





Conclusion

Proton TMDPDFs from Lattice QCD

- We have presented the **first simultaneous** lattice-QCD determination of full isovector **proton** light-cone TMDPDFs in the **unpolarized, helicity, and transversity** channels;
- Although the three channels have distinct spin structures and different normalizations, their **non-perturbative transverse structures** are similar within the current uncertainties;
- After renormalization, the large- b_T behavior of TMD PDFs is dominated by the **intrinsic soft function**;
- After combining the **short distance OPE** and large distance extrapolation, the k_T -space TMDPDFs provides a spin-dependent three-dimensional proton tomography;
- In future work, we will generalize to **larger P_z and multi-ensemble analysis**.



LaMET-Agent: For Data Analysis in LaMET

Why we need Agents?

- How to promote a method / framework? - Make it as user-friendly as possible!
- Agent is a new format to promote and present a method / framework.

Why we need LaMET-Agent?

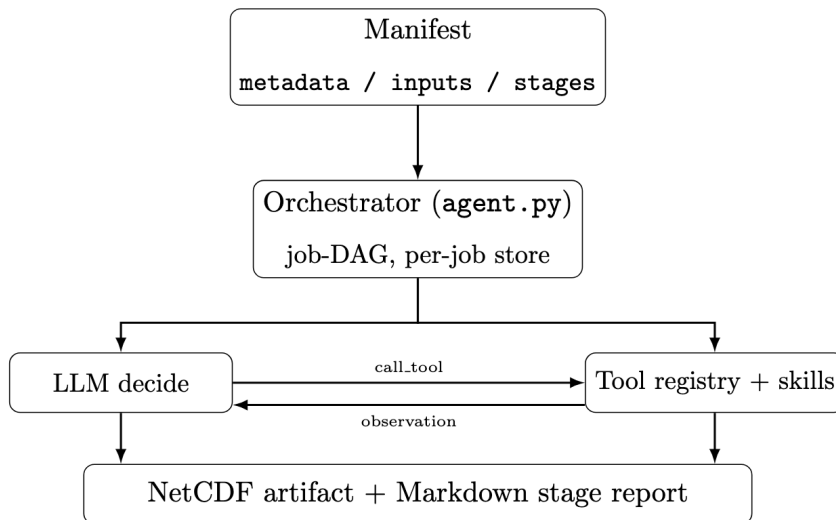
- **For researchers new to LaMET:** the agent makes LaMET data analysis accessible and helps users understand the **correct** analysis workflow and methodology;
- **For experienced LaMET users:** the agent automates routine analysis tasks, accelerates the workflow, and allowing researchers to focus on physics instead of implementation;
- LaMET-Agent provides a standard, well-organized, understandable workflow.



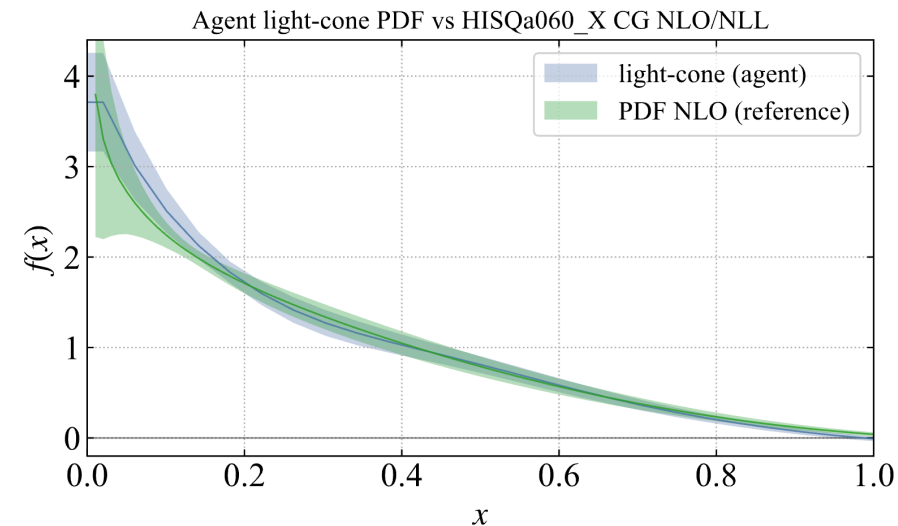
LaMET-Agent: For Data Analysis in LaMET

A delicate combination of deterministic pipelines and LLM decisions

- Six stages: Correlator Analysis, Renormalization, Fourier Transform, Perturbative Matching, Physical Extrapolation, Review;
- Single-run workflow for multi-ensemble analyses; unified NetCDF-based data management; stage reports for transparency; easily extensible to new observables and analysis techniques.



Reproducing the Pion PDF with LaMET Agent



X. Gao, W. Y. Liu and Y. Zhao, PRD 109 (2024)



Thank you!

Jinchen He @ DPF Meeting 2026

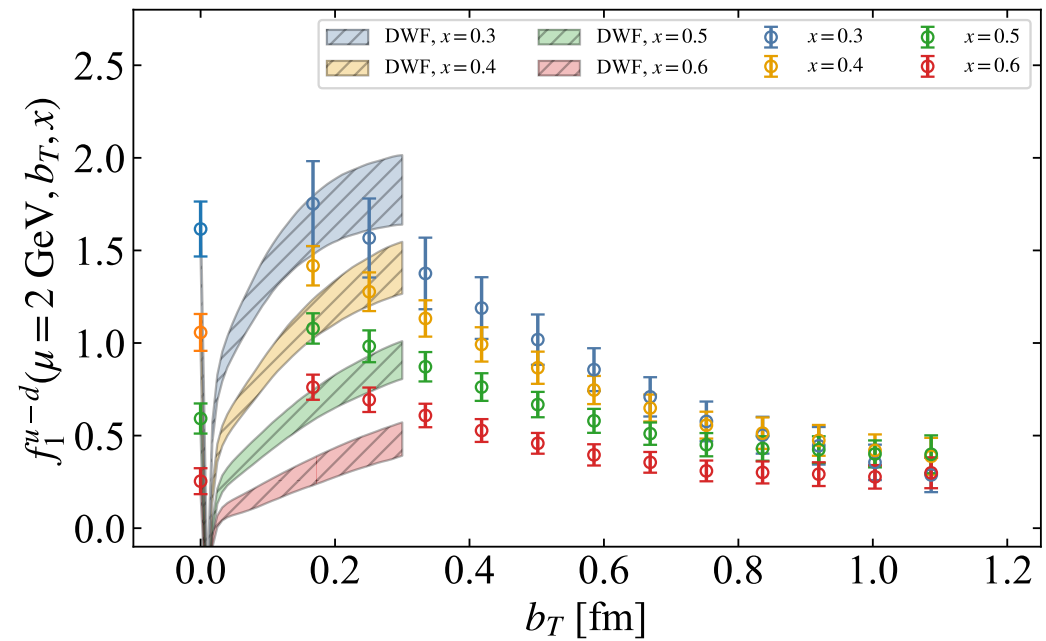
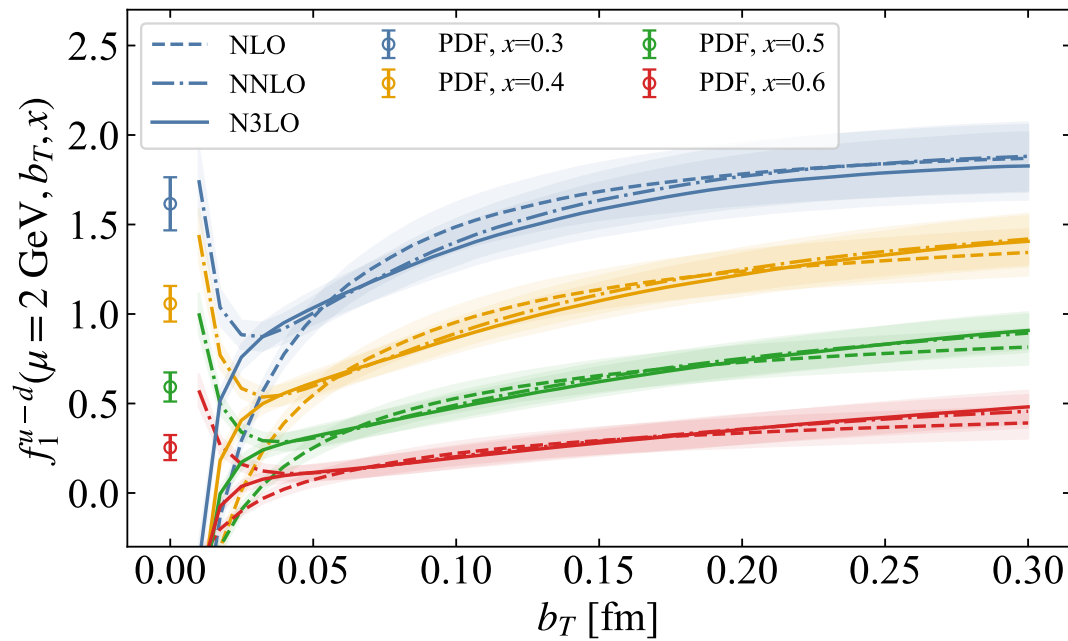


Short distance OPE

TMD PDFs admit an OPE onto the corresponding collinear PDFs at perturbative b_T

- The intrinsic soft function cannot be directly calculated on lattice because of two light-like

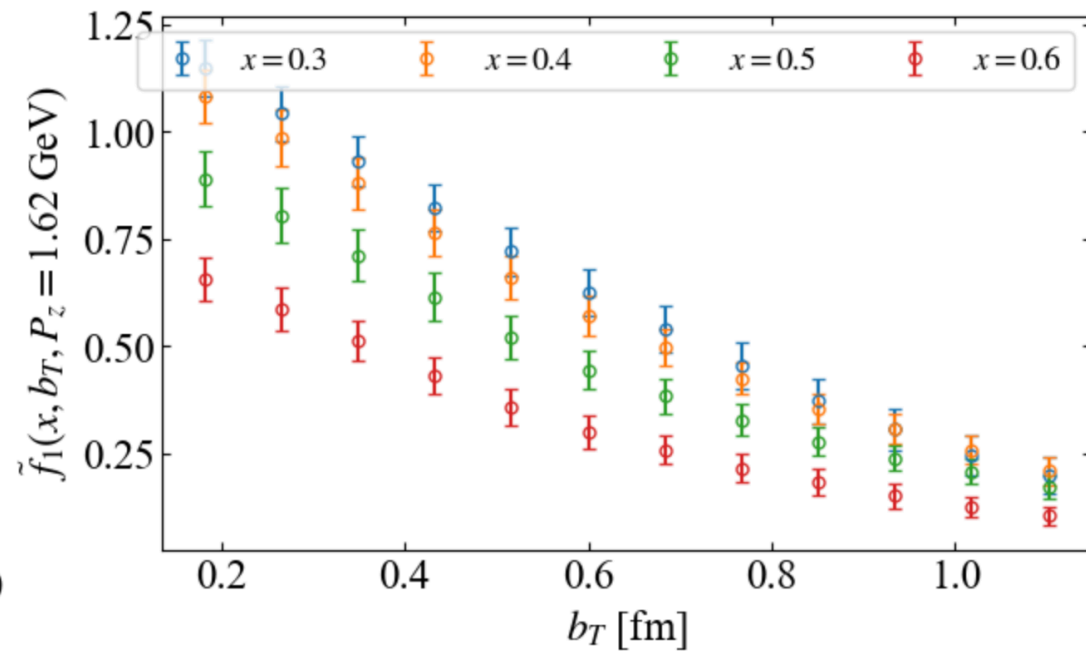
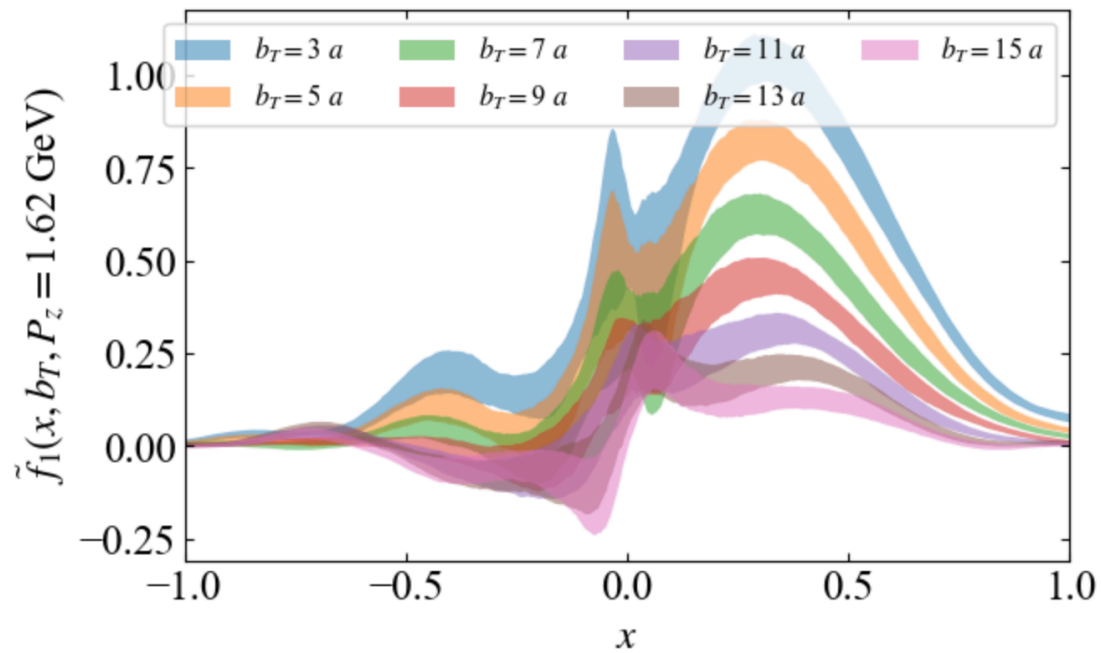
$$f_{\Gamma}(x, b_T; \mu, \zeta) = C_{\Gamma}(x, b_T; \mu, \zeta) \otimes f_{\Gamma}^{\text{coll}}(x, \mu) + \mathcal{O}(b_T^2 \Lambda_{\text{QCD}}^2)$$





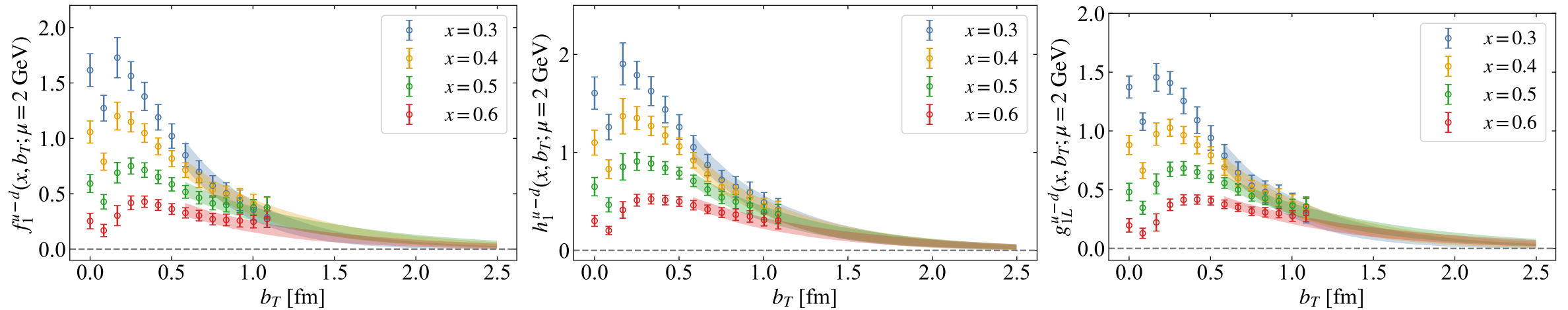
bT Dependence of Bare qTMD

What if before renormalization?



Extrapolation in Larger bT Region

Extrapolation using the exponential form and loose prior from soft functions



Soft Functions in TMD Factorization

$$\frac{d\sigma}{dQdYd^2\mathbf{q}_T} = \left(\frac{d\sigma^W}{dQdYd^2\mathbf{q}_T} + \frac{d\sigma^Y}{dQdYd^2\mathbf{q}_T} \right) \left[1 + \mathcal{O}\left(\frac{\Lambda_{\text{QCD}}^2}{Q^2}\right) \right].$$

$$\frac{d\sigma^W}{dQdYd^2\mathbf{q}_T} = \sum_{\text{flavors } i} H_{i\bar{i}}(Q^2, \mu) \int d^2\mathbf{b}_T e^{i\mathbf{b}_T \cdot \mathbf{q}_T} \tilde{f}_{i/p}(x_a, \mathbf{b}_T, \mu, \zeta_a) \tilde{f}_{\bar{i}/p}(x_b, \mathbf{b}_T, \mu, \zeta_b) \quad (2.29a)$$

$$= \sum_{\text{flavors } i} H_{i\bar{i}}(Q^2, \mu) \int d^2\mathbf{b}_T e^{i\mathbf{b}_T \cdot \mathbf{q}_T} \tilde{B}_{i/p}(x_a, \mathbf{b}_T, \mu, \zeta_a/v^2) \tilde{B}_{\bar{i}/p}(x_b, \mathbf{b}_T, \mu, \zeta_b/v^2) \times \tilde{S}_{n_a n_b}(b_T, \mu, \nu). \quad (2.29b)$$

$$\tilde{f}_{i/p}(x, \mathbf{b}_T, \mu, \zeta) = \tilde{B}_{i/p}(x, \mathbf{b}_T, \mu, \zeta/v^2) \sqrt{\tilde{S}_{n_a n_b}(b_T, \mu, \nu)}, \quad (2.32)$$

$$\begin{aligned} \tilde{B}_{i/p}(x, \mathbf{b}_T, \mu, \zeta/v^2) &= \lim_{\substack{\epsilon \rightarrow 0 \\ \tau \rightarrow 0}} \tilde{Z}_B^i(b_T, \mu, \nu, \epsilon, \tau, xP^+) \tilde{B}_{i/p}^0(x, \mathbf{b}_T, \epsilon, \tau, xP^+) \\ &= \lim_{\substack{\epsilon \rightarrow 0 \\ \tau \rightarrow 0}} \tilde{Z}_B^i(b_T, \mu, \nu, \epsilon, \tau, xP^+) \frac{\tilde{f}_{i/p}^{0(u)}(x, \mathbf{b}_T, \epsilon, \tau, xP^+)}{\tilde{S}_{n_a n_b}^{0\text{subt}}(b_T, \epsilon, \tau)}, \end{aligned} \quad (2.34)$$

$$\tilde{S}_{n_a n_b}(b_T, \mu, \nu) = \lim_{\substack{\epsilon \rightarrow 0 \\ \tau \rightarrow 0}} \tilde{Z}_S(b_T, \mu, \nu, \epsilon, \tau) \tilde{S}_{n_a n_b}^0(b_T, \epsilon, \tau). \quad (2.35)$$

$$\begin{aligned} \tilde{f}_\Gamma(x, b_\perp, P^z; \mu) &= H_f(x, P^z; \mu) B(x, b_\perp, xP^+; \mu, \nu) \\ &\times S_C^0(b_\perp; \mu, \nu), \end{aligned} \quad (3)$$

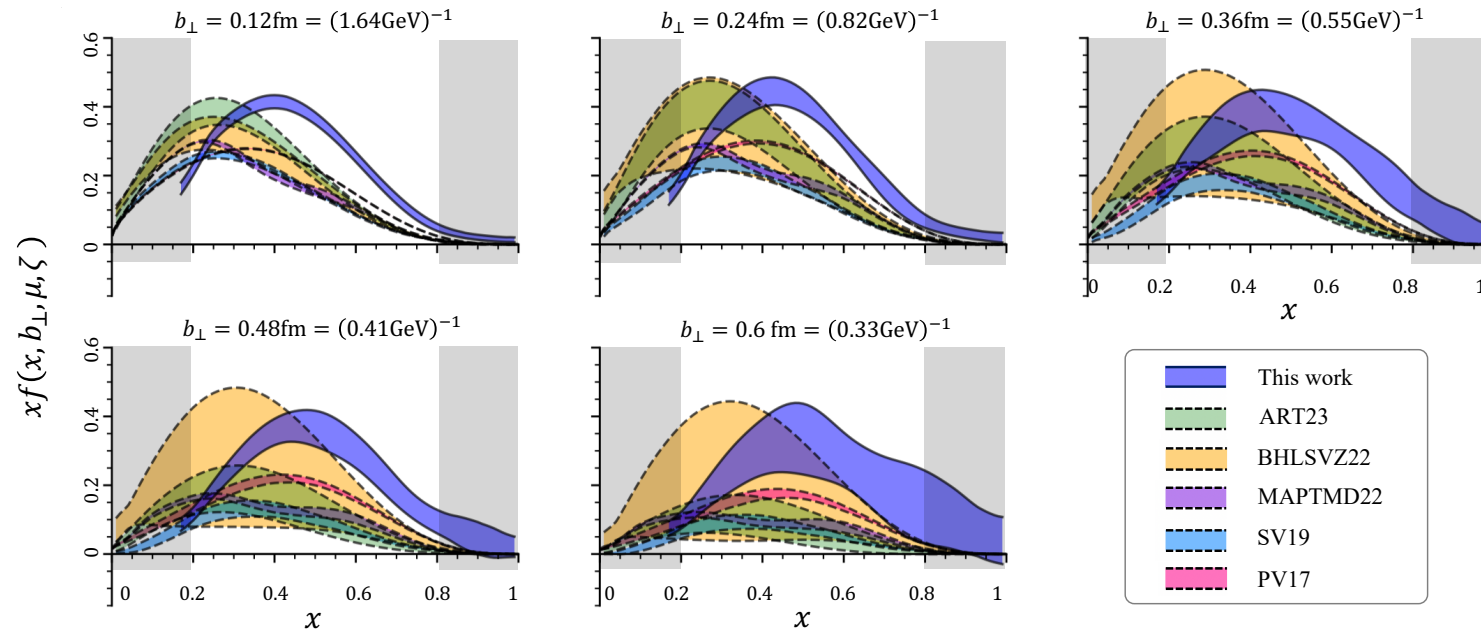
$$\begin{aligned} \sqrt{S_I(b_\perp, y_n; \mu)} \cdot \tilde{f}_\Gamma(x, b_\perp, P^z; \mu) \\ = f(x, b_\perp; \mu, \zeta) H_f(x, P^z; \mu) + \text{p.c.}, \end{aligned} \quad (5)$$

- Zero-bin subtracted beam function has no soft contribution;
- S_C^0 is the quasi-soft function with zero-bin contribution;
- The intrinsic soft function S_I is the combination of subtracted soft function and the quasi-soft function



Unpolarized Proton TMDPDFs

First lattice QCD calculation of x-dependent proton TMDPDFs



♦ $a = 0.12$ fm

♦ $P_{\text{max}}^z = 2.58$ GeV

♦ Physical limit of m_{π}^{val}

♦ N3LL matching
(Next-to-next-to-next-to-leading logarithmic)

♦ NLO soft function
(Next-to-leading order)

JH, et al. (LPC), Phys.Rev.D 109 (2024)

- Systematics are estimated and included in the final results;
- Constrained by the limited computing resources, it is hard to implement on a finer lattice;
- Due to the signal-to-noise ratio (SNR), it is hard to probe the large b_T region.



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